

Gideon Bradburd

June 8, 2026

3042 Biological Sciences Building
Department of Ecology & Evolutionary Biology
University of Michigan

email: bradburd@umich.edu
website: genescape.org
software: github.com/gbradburd

EDUCATION & APPOINTMENTS

Assoc. Professor	University of Michigan Department of Ecology and Evolutionary Biology	2025 - present
Asst. Professor	University of Michigan Department of Ecology and Evolutionary Biology	2022 - present
Asst. Professor	Michigan State University Department of Integrative Biology Ecology, Evolution, and Behavior Program fixed-term (2016-17); tenure-stream (2017-2022)	2016 - 2022
Postdoc	University of California, Berkeley Museum of Vertebrate Zoology Department of Env. Sci. Pol. and Mgmt Advisors: Michael Nachman, Bree Rosenblum	2015 - 2016
PhD	University of California, Davis Population Biology Graduate Group Department of Ecology and Evolutionary Biology Advisors: Graham Coop, Brad Shaffer	2009 - 2015
Bachelor of Science	Yale University Ecology and Evolutionary Biology Honors Thesis Advisor: Tom Near	2004 - 2008

LEAVES & MODIFIED DUTIES

2020	Second child born in Aug, 6 weeks parental leave, teaching release Spring Semester 2021
2016	First child born in November, 3 weeks parental leave, one-year tenure clock extension

PUBLICATIONS

*advisees (e.g., students/postdocs)

Preprints, In Review, and In Revision

- *Talbot, CA, GS Bradburd. 2025. ARGscape: A modular, interactive tool for manipulation of spatiotemporal ancestral recombination graphs. [arXiv](https://arxiv.org/abs/2505.12345).
- *Toczydlowski, RH, RS Brennan, ED Crandall, JL Kelley, JM Pringle, C Riginos, JP Wares, GS Bradburd. 2025. Drivers of genetic diversity across the marine tree of life. [bioRxiv](https://arxiv.org/abs/2505.12345).
- *Hancock, ZB, GS Bradburd. 2025. sandworm: an R package for estimating Wright's neighborhood size and species-level genetic diversity. [bioRxiv](https://arxiv.org/abs/2505.12345).

Published and Accepted

34. *Hancock, ZB, NE Adams, PL Wood, GS Bradburd. **2025**. Asymmetric gene flow maintains an abundant range center. **American Naturalist**. (accepted). [bioRxiv](#).
33. Smith, SA, JB Pease, T Carruthers, GS Bradburd, IB Huegele, GW Stull, WN Weaver, Y Yang, TS Yi, JM Beaulieu. **2025**. Longevity in plants impacts phylogenetic and population dynamics. **New Phytologist** (in press).
32. Clark, MI, ET Hileman, JA Moore, LJ Faust, RE Junge, BN Reid, DR Bradke, GS Bradburd, SW Fitzpatrick. **2025**. Inbreeding reduces fitness in spatially structured populations of a threatened rattlesnake, **Proceedings of the National Academy of Sciences** 122 (34) e2501745122.
31. Buysse SF, SG Pérez, JR Puzey, A Garrison, GS Bradburd, CG Oakley, SJ Tonsor, FX Picó, EB Josephs, JK Conner. **2025**. Evaluating the Roles of Drift and Selection in Trait Loss along an Elevational Gradient. **Evolution**. qpa078.
30. *Grundler, MC, J Terhorst, GS Bradburd. **2025**. A geographic history of human genetic ancestry. **Science**. 387,1391-1397.
29. *Clark, MI, SW Fitzpatrick, GS Bradburd. **2024**. Pitfalls and windfalls of detecting demographic declines using population genetics in long-lived species. **Evolutionary Applications**. 17 (7), e13754.
28. Cocciardi JM, AM Hoffman, A Waananen, DL Des Marais, D Moeller, D Gamba, D Alvarado-Serrano, E Boehm, E Kottler, G Bradburd, H Branch, I Borokini, J Cavender-Bares, J Lau, J Anderson, J Jaros, K Toll, K Whitney, L Bolin, L Brudvig, M Ungerer, M Vahsen, M Blumstein, M Smith, M Howard, M Menon, NP Hanan, N Kooyers, R Shaw, S Sheth, S Wadgyamar, T Mozdzer, T Juenger, T Chen, ML Avolio. **2024**. The value of long-term ecological research for evolutionary insights. **Nature Ecology & Evolution**. 8, 1584–1592.
27. *Hancock, Z, *RH Toczydlowski, GS Bradburd. **2024**. A spatial approach to jointly estimate Wright's neighborhood size and long-term effective population size. **Genetics**. 227 (4).
26. *Lewanski, AL, *MC Grundler, GS Bradburd. **2024**. The era of the ARG: An introduction to ancestral recombination graphs and their significance in empirical evolutionary genomics. **PLoS Genetics**. 20(1): e1011110.
25. *Week, B and GS Bradburd. **2024**. Host-parasite coevolution in continuous space leads to variation in local adaptation across spatial scales. **American Naturalist** 203:1, 43-54.
24. Mathur, S, A Mason, GS Bradburd, HL Gibbs. **2023**. Functional genomic diversity is correlated with neutral genomic diversity in populations of an endangered rattlesnake. **Proceedings of the National Academy of Sciences** 120 (43).
23. Crandall, ED, *RH Toczydlowski, L Liggins, AE Holmes, M Ghoojaei, MR Gaither, BE Wham, AL Pritt, C Noble, TJ Anderson, RL Barton, JT Berg, SG Beskid, A Delgado, E Farrell, N Himmelsbach, SR Queeno, T Trinh, C A Weyand, A Bentley, J Deck, C Riginos, GS Bradburd, RJ Toonen. **2023**. The importance of timely metadata curation to the global surveillance of genetic diversity. **Conservation Biology** 37, e14061.
22. Martin, BS, GS Bradburd, LJ Harmon, MG Weber. **2022**. Modeling the evolution of rates of continuous trait evolution. **Systematic Biology** 72 (3), 590-605.
21. *Clark, M, GS Bradburd, M Akopyan, A Vega, E Rosenblum, J Robertson. **2022**. Genetic isolation by distance underlies color pattern divergence in red-eyed treefrogs (*Agalychnis callidryas*). **Molecular Ecology**. 31: 1666-1681.

20. *Toczydlowski, RH, L Liggins, MR Gaither, TJ Anderson, RL Barton, JT Berg, SG Beskid, B Davis, A Delgado, E Farrell, M Ghoojajei, N Himmelsbach, AE Holmes, SR Queeno, T Trinh, CA Weyand, GS Bradburd, C Riginos, RJ Toonen, ED Crandall. **2021**. Poor data stewardship will hinder global genetic diversity surveillance. **Proceedings of the National Academy of Sciences** 118 (34).
19. *Puckett, E, S Murphy, GS Bradburd. **2021**. Phylogeographic analysis delimits three evolutionary significant units of least chipmunks in North America and identifies unique genetic diversity within the imperiled Peñasco population. **Ecology and Evolution** 11: 12114-12128.
18. Schweizer, RS, MR Jones, GS Bradburd, JF Storz, N Senner, C Wolf, ZA Cheviron. **2021**. Broad concordance in the spatial distribution of adaptive and neutral genetic variation along an elevational cline in deer mice. **Molecular Biology and Evolution** 38 (10), 4286-4300.
17. *Hancock, ZB, ES Lehmborg, GS Bradburd. **2021**. Neo-darwinism still haunts evolutionary theory: A modern perspective on Charlesworth, Lande, and Slatkin (1982). **Evolution** 75: 1244-1255.
16. Rothstein, A, R Knapp, GS Bradburd, D Boiano, EB Rosenblum. **2020**. Stepping into the past to conserve the future: archived skin swabs from extant and extinct populations inform genetic management of an endangered amphibian. **Molecular Ecology** 29 (14): 2598-2611.
15. Fitzpatrick, SW GS Bradburd, CT Kremer, PE Salerno, LM Angeloni, WC Funk. **2020**. Genomic and fitness consequences of genetic rescue in wild populations. **Current Biology** 30: 1-6.
14. Schweizer, RM, JP Velotta, CM Ivy, MR Jones, SM Muir, GS Bradburd, JF Storz, GR Scott, ZA Cheviron. **2019**. Physiological and genomic evidence that a transcription factor contributes to adaptive cardiovascular function in high-altitude deer mice. **PLoS Genetics** 15 (11), e1008420.
13. Bradburd, GS and PL Ralph. **2019**. Spatial population genetics: It's about time. **Annual Reviews in Ecology, Evolution, and Systematics** 50: 427-449.
12. Grieneisen, LE, MJE Charpentier, SC Alberts, R Blekman, GS Bradburd, J Tung, EA Archie. **2019**. Genes, geology, and germs: gut microbiota across a primate hybrid zone are explained by site soil properties, not host species. **Proceedings of the Royal Society B** 286: 20190431.
11. Bradburd, GS, GM Coop, and PL Ralph. **2018**. Inferring continuous and discrete population genetic structure across space. **Genetics** 210: 33-52.
10. Weber, JN, GS Bradburd, YE Stuart, WE Stutz, DI Bolnick. **2017**. Partitioning the effects of isolation by distance, environment, and physical barriers on genomic divergence between parapatric threespine stickleback. **Evolution** 71: 342-56.
9. Hoban, S, JL Kelley, KE Lotterhos, MF Antolin, GS Bradburd, DB Lowry, ML Poss, LK Reed, A Storfer, MC Whitlock. **2016**. Finding the genomic basis of local adaptation in non-model organisms: pitfalls, practical solutions, and future directions. **American Naturalist** 188: 379-397.
8. Bradburd, GS, PL Ralph, GM Coop. **2016**. A spatial framework for understanding population structure and admixture. **PLoS Genetics** 12: e1005703.
7. Hammock, BG, S Lesmeister, I Flores, GS Bradburd, FH Hammock, SJ Teh. **2016**. Low food availability narrows the tolerance of the copepod *Eurytemora affinis* to salinity, but not to temperature. **Estuaries and Coasts**. 39: 189-200.
6. Agrawal, AA, AP Hastings, GS Bradburd, EC Woods, T Züst, J Harvey, T Bukovinszky. **2015**. Evolution of plant growth and defense in a continental introduction. **American Naturalist** 186: 1-15.
5. Wang, IJ† and GS Bradburd†. **2014**. Isolation by environment. **Molecular Ecology** 23: 5649-5662.
(†co-first authors)

4. Bradburd, GS, PL Ralph, and GM Coop. **2013**. Disentangling the effects of geographic and ecological isolation on genetic differentiation. **Evolution** 67: 3258-3273.
3. Rejmanek, D, P Freycon, GS Bradburd, J Dinstell, and J Foley. **2013**. Unique strains of *Anaplasma phagocytophilum* segregate among diverse questing and non-questing *Ixodes* tick species in the western United States. **Ticks and Tick-borne Diseases** 4: 482-487.
2. Rejmanek, D, GS Bradburd, and J Foley. **2012**. Molecular characterization reveals distinct genospecies of *Anaplasma phagocytophilum* from diverse North American hosts. **Journal of Medical Microbiology** 61: 204-212.
1. Near, TJ, CM Bossu, GS Bradburd, RL Carlson, RC Harrington, PR Hollingsworth Jr., BP Keck, DA Etnier. **2011**. Phylogeny and temporal diversification of darters (*Percidae: Etheostomatinae*). **Systematic Biology** 60: 565-595.

SOFTWARE

- Talbot, CA, GS Bradburd (2025). **ARGscape**. An interactive platform for simulating, analyzing, and visualizing Ancestral Recombination Graphs across space and time. Python package vo.3.0. <https://www.argscape.com/>
- Hancock, ZB, GS Bradburd (2025). **sandworm**. A method for inferring Wright's neighborhood size and deep population diversity from spatial genomic data. R package vo.1.0. <https://github.com/zachbhancock/sandworm>
- Grundler, M, J Terhorst, GS Bradburd (2024). **GAIA**. A method for inferring the geographic locations of shared genetic ancestors in a tree sequence. R package v1.0. <https://github.com/blueraleigh/gaia>
- Bradburd, GS (2018). **conStruct**. A genetic clustering method that accounts for isolation by distance. R package v1.0.6. <https://cran.r-project.org/web/packages/conStruct/index.html>
- Bradburd, GS (2016). **SpaceMix**. A program for estimating geogenetic maps from allele frequency data. R package vo.13. <https://github.com/gbradbud/SpaceMix>
- Bradburd, GS (2013). **BEDASSLE**. Bayesian Estimation of Differentiation in Alleles by Spatial Structure and Local Ecology. R package v1.6.1. <https://cran.r-project.org/web/packages/BEDASSLE>

FUNDING

External

NIH NIGMS MIRA (R35) Award Renewal# R35GM137919 (2026-20230)	\$2,337,560
Incorporating geography into statistical methods for analysis of population genomic DNA (PI: G Bradburd)	
NSF Division of Environmental Biology Award# 2223962 (2022-2026)	\$1,536,017
The interaction of selection, pleiotropy, and drift in phenotypic evolution (PI: J Conner; co-PIs: G Bradburd, R Last, T Juenger)	
Bradburd lab budget: \$210,974.4	
NIH NIGMS Early Stage Investigator MIRA (R35) Award# R35GM137919 (2020-2025)	\$1,849,678
Incorporating geography into statistical methods for analysis of population genomic DNA (PI: G Bradburd)	
NSF Bridging Ecology & Evolution Award# 2016569 (2020-2024)	\$920,700
Testing eco-evolutionary effects of genetic drift and gene flow in stressful environments (PI: S Fitzpatrick; co-PIs: B Rothermel, G Bradburd)	
Bradburd lab budget:\$157,590	

NSF Dimensions of Biodiversity (2017-2022)	\$1,118,520
Diversification of sensory systems in novel habitat (PI: J Boughman; co-PIs: G Bradburd, H Hofmann, J Keagy, D Stenkamp) Bradburd lab budget: \$158,851	
NSF Doctoral Dissertation Improvement Grant (DDIG) (2014-15)	\$13,918
The effect of intraspecific host variation on the structure of parasite populations (PI: G Bradburd; co-PI: G Coop)	
<u>Internal</u>	
Biodiversity Exploration Fund (2025-2026)	\$11,586
University of Michigan Museum of Natural History (PI: G Bradburd. co-PI: Stephen Smith)	
MSU Strategic Partnership Grant (2021-2022)	\$400,000
The Institute for Biodiversity, Ecology, Evolution, and Macrosystems (IBEEM) (PI: PL Zarnetske. co-PIs: KS Cheruvellil, E Zipkin, KM Dahlin, G Bradburd, J Robinson)	
Beacon - NSF Center for the Study of Evolution in Action (2016-2021)	\$127,579
Evolution of sensory systems in response to loss of visual information (PI: J Boughman, co-PIs: G Bradburd, H Hofmann, J Keagy, D Stenkamp)	

MENTORSHIP & ADVISING

Postdocs

Seth Temple (Schmidt AI in Science Fellow)	2024-present
Mike Grundler	2022-2025
<i>current position:</i> Assistant Professor, Southern Illinois University	
Zach Hancock	2021-2025
<i>current position:</i> Assistant Professor, Augusta University	
Leonard Jones (NSF Postdoctoral Research Fellowship in Biology)	2021-2024
<i>current position:</i> Bondermann Fellow, University of Washington	
Phil Gerrish	2022-2024
<i>current position:</i> Researcher, Los Alamos National Lab	
Bob Week	2020-2022
<i>current position:</i> Postdoctoral Researcher, University of Oregon	
Rachel Toczydlowski	2019-2022
<i>current position:</i> Research Landscape Geneticist, US Forest Service	
Matteo Tomasini	2019-2021
<i>current position:</i> Research Software Engineer, University of Gothenburg	
Kelsey Yule	2018-2019
<i>current position:</i> Project Manager of the NEON Biorepository, Arizona State University	
Emily Puckett	2017 - 2018
<i>current position:</i> Asst. Professor of Biological Sciences, U. Tennessee, Memphis	

PhD Students

Rachel Gutner	2025-present
Alex Lewanski (DoD National Defense Science and Engineering Graduate Fellowship)	2022-present
Meaghan Clark	2019-2024
<i>current position:</i> NSF Postdoctoral Research Fellow, UC Santa Cruz	

Master's Students

Ivana Barnes	2024-present
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Bioinformaticians/Technicians

Nicole Adams	2022-present
Chris Talbot	2024-2025

Graduate student committee service

Current graduate student committees

Yeishmary Soto Muñiz	PhD, EEB, University of Michigan
Greety Wuysang	Masters, Biology, James Madison University
Denise Meier	PhD, EEB, University of Michigan
Jiatong Liang	PhD, Statistics, University of Michigan
Kaira Schaefer	Masters, EEB, University of Michigan
Mikayla Creek	Masters, Biology, James Madison University
Max Wytinski	PhD, EEB, University of Michigan
Nepsis García	PhD, EEB, University of Michigan
Eric Bastien	PhD, EEB, University of Michigan
Gayathri Venkatraman	PhD, EEB, University of Michigan
Kiran Kumar	PhD, Biostatistics, University of Michigan
Tristan Schramer	PhD, EEB, University of Michigan
Teresa Sauer	PhD, EEB, University of Michigan
Yu Kai Tan	PhD, EEB, University of Michigan
Sofia Maria Backlund	PhD, Institute of Science and Technology (Vienna)
Jimmy Bingman	PhD, Plant Biology, MSU
Matheus Januario Lopes de Sousa	PhD, EEB, University of Michigan

Past graduate student committees

Bruce Martin	PhD 2024, Plant Biology, MSU
Enes Dilber	PhD 2023, Statistics, University of Michigan
Miranda Wade	PhD 2023, Integrative Biology, MSU
Sara Hugentobler	PhD 2023, Integrative Biology, MSU
Kyle Teller	PhD 2022, Integrated Applied Math, University of New Hampshire
Viviana Ortiz Londono	PhD 2022, Plant Pathology, MSU
Ava Garrison	PhD 2022, Plant Biology, MSU
Michael Foisy	MSc 2020, Plant Biology, MSU
Dr. Emily Dolson	PhD 2019, Computer Science and Engineering, MSU

Undergraduates

Isla Roy	2025 - present
Ben Carlson	2024 - present
Guoxiang (Jerry) Guo	2024 - 2025
Lindsay Guare (MSU)	2017 - 2019
Sarah Frocillo (MSU)	2017 - 2018
Nick Stants (MSU)	2017 - 2018

HONORS, AWARDS, & FELLOWSHIPS

Fitch Award Finalist , Society for Molecular Biology and Evolution (\$2000)	2015
Hamilton Award Finalist , Society for the Study of Evolution (\$500)	2015
XSEDE SuperComputing Resource Allocation , XSEDE (50,000 SUs)	2014-05
Center for Population Biology Research Fellowship CPB, UC Davis (\$4,000)	2011-13
NSF Graduate Research Fellow National Science Foundation (\$90,000)	2010-12
Graduate Scholars Fellowship in Population Biology, UC Davis (\$30,000)	2009-10

TEACHING

University of Michigan

Introduction to Statistical Model Building in R (EEB 429)	Instructor	WN23-25
<i>Undergraduate course (20-40 students/yr); introduction to coding and statistics in R.</i>		
Evolution (EEB 390)	Instructor	WN23,FA24
<i>Undergraduate course (100-150 students/yr); introduction to evolutionary biology.</i>		

Michigan State University

Evolutionary Biology (IBio 849)	Instructor	SP18,19,20,22
<i>Graduate-level course (20-40 students/yr); introduction to evolutionary biology.</i>		
Intro to Statistical Methods in Ecology/Evolution (IBio 830)	Instructor	FA17,18,19,20
<i>Graduate-level course (40-60 students/yr); introduction to coding and statistics in R.</i>		
Evolution Discussion Group (IBio 895)	Instructor	SP17,18,19,20
<i>Graduate student-led discussion group on a rotating technical topic in evolutionary biology.</i>		

SERVICE & LEADERSHIP

National/International/Society Service

Organizer SMBE Symposium: "Spatial population genetics: where are we now?"	2024
Organizing Committee Midwest Population Genetics Conference	2016-present
Ad hoc Associate Editor , PLoS Genetics	2025
Conference Ally Society for Molecular Biology and Evolution	2024
Steering Committee NSF EVO-LTER Award: Leveraging long-term ecological research in grasslands: facilitating collaborations between ecologists and evolutionary biologists	2021-present
Course Instructor UCLA/La Kretz Center Workshop in Conservation Genomics	2014, 16, 19
Judge Hamilton Award (Society for the Study of Evolution)	2017, 19
Ad hoc Grant Reviewer : NSERC, ERC, NSF, Austrian Science Fund, UKRI	
Reviewer including: PNAS; Ecology Letters; Evolution Letters; Genetics; Evolution; American Naturalist; Systematic Biology; Molecular Biology and Evolution; PLoS Genetics; Nature Genetics; Nature Communications; Molecular Ecology; Proceedings of the Royal Society B; American Journal of Human Genetics; Heredity; Molecular Ecology Resources; Methods in Ecology and Evolution; Molecular Phylogenetics and Evolution; New Phytologist; Bioinformatics; Ecology	

Institutional Service

University of Michigan

Director, Frontiers Master's Program	2025-present
Nominating Committee (College of LSA)	2024
Executive Committee (EEB Dept.)	2023-present
Executive Committee, UM Genome Science Training Program	2023-present
Collegiate Fellows Faculty Search Committee (EEB Dept.)	2023
Founder, Exploring Careers Beyond Academia with Lunch (EEB Dept.)	2023-present
Ad hoc Faculty Search Committee (EEB Dept.)	2023,2024
Seminar Committee (EEB Dept.)	2022-2023

Michigan State University

Diversity, Equity, and Inclusion Committee (Dept. Integrative Biology)	2019-2022
Co-founder, Diversity, Equity, and Inclusion Reading Group (Dept. Integrative Biology)	2019-2022
Co-chair, Presidential Postdoc Fellowship Committee (EEB Program)	2020-2022

Seminar Committee (Dept. Integrative Biology)	2020-2022
Chair Search Committee (Dept. Integrative Biology)	2020-2022
Seminar Committee (EEB Program)	2019-2020
Director Search Committee (EEB Program)	2019-2020
<i>Ad hoc</i> Space Committee (EEB Program)	2019-2020
Strategic Hiring Planning Committee (Dept. Integrative Biology)	2018-2019

INVITED SEMINARS

University of Otago	Department of Zoology	2026
University of Michigan	University of Michigan Biological Station	2024
European Society for Evolutionary Biology	Integration of Speciation Research	2024
Louisiana State University	Dept. of Biological Sciences	2022
University of Michigan	Dept. of Ecology and Evolutionary Biology	2022
Institute of Science and Technology Austria	EvoLunch	2021
North Dakota State University	Dept. of Biological Sciences	2021
University of St. Andrews	Centre of Ecol. and Env. Modelling	2021
Cornell University	Dept. of Computational Biology	2020
Western Michigan University	Dept. of Biological Sciences	2020
Ohio State University	Dept. of Evolution, Ecology, Organismal Bio.	2019
University of Kentucky	Dept. of Biology	2019
Ecological Society of America	Landscape Genetics (student-organized)	2018
Indiana University	Dept. of Biology/Precision Health Initiative	2018
Penn State University	Ecology Seminar Series	2018
University of Toronto (St. George)	Dept. of Ecology and Evolutionary Biology	2018
University of Toronto (Mississauga)	Dept. of Biology	2017
University of Chicago	Dept. of Ecology and Evolution	2017
Michigan State University	Kellogg Biological Station	2017
University of Michigan	Center for Statistical Genetics	2016
University of Wyoming	Dept. of Botany	2016
University of Colorado Boulder	Dept. Ecology and Evolutionary Biology	2016
University of San Francisco	Dept. of Biology	2016
University of California Berkeley	Museum of Vertebrate Zoology	2015
Michigan State University	Ecology, Evolution, and Behavior	2015
Stanford University	Dept. Genetics and Biology	2015
University of Southern California	Molecular and Computational Biology	2014
University of Montana	Division of Biological Sciences	2014
University of Minnesota	Dept. of Plant Biology	2014
Sonoma State University	Biology Dept.	2014
University of Texas, Austin	Dept. Integrative Biology	2013

CONTRIBUTED PRESENTATIONS

Society for the Study of Evolution	2012-13,15,17,19,23
Population, Evolutionary, Quantitative Genetics Meeting	2018, 2022
Midwestern Population Genetics Meeting	2016
American Society of Naturalists	2016,18
Society for Molecular Biology and Evolution	2015
Bay Area Population Genomics (BAPG)	2014